METHODOLOGICAL NOTE · MERGE DOSE ESCALATION

# Treating a dose-limiting toxicity as a probability rather than a yes or no

A single contested toxicity currently costs the design as much as an unambiguous one. This note asks whether attribution should enter the model as a fraction, reviews what already exists in the literature, and quantifies what it would change.

SIMULATIONS **1000** PER CELL   ACUTE TARGET **0.20**   SUBACUTE TARGET **0.33**

CONFIGURATION **NO EWOC, NO BURN-IN**

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## 1 The problem with a binary attribution

The sixth patient treated at L1 experienced an event assessed as “possible” — not unrelated, not definite. The design has no way to express that. It must record either a 1 or a 0, and the two answers lead to very different trials.

This is not a peculiarity of our study. Adverse event attribution is routinely collected on a five-tier scale — unrelated, unlikely, possible, probable, definite — and then collapsed to a binary for the purpose of dose escalation. The graded judgement the clinicians actually made is discarded at exactly the point where it would matter most.

There is good evidence that the binary is unreliable. A review of randomised phase III trials found that **half of the adverse events reported on placebo arms were recorded as drug related**, and that among patients with a repeated event, **36% changed attribution over time**. Forcing a confident 0 or 1 out of a genuinely uncertain judgement does not remove the uncertainty; it hides it.

> Radiotherapy makes this sharper still. The tumour itself produces events that look like treatment toxicity — as in this case, where disease progression and irradiation are competing explanations for the same finding.

## 2 This has been done before

The idea of a non-binary toxicity endpoint is established, which is useful: it means the approach can be cited rather than invented.

### Non-binary toxicity by severity

The quasi-CRM of Yuan, Chappell and Bailey (2007) replaces the binary outcome with an equivalent toxicity score and fits it through a quasi-Bernoulli likelihood — grade 2 counts as 0.5, grade 3 as 1, grade 4 as 1.5. The machinery for a fractional DLT has therefore been in routine use for close to twenty years, including extensions to drug-combination designs.

### Non-binary toxicity by attribution

Closer to the present question, two published designs handle attribution itself as uncertain: *Phase I designs that allow for uncertainty in the attribution of adverse events* (JRSS-C), and *A Bayesian dose-finding design for outcomes evaluated with uncertainty* (2021). The latter is almost exactly the proposal here — for some patients the physician records a probability of DLT rather than a binary, and the posterior is obtained by data augmentation over the unobserved binary outcome, with the CRM as the worked example.

So the proposal is not novel in its mechanics. What is unexplored is the part specific to radiotherapy, discussed in section 5.

## 3 Discounting the event is not the same as discounting the patient

This distinction is easy to miss and changes the answer. Our earlier analysis down-weighted the historical data the wrong way.

A TITE-CRM likelihood accumulates two quantities per dose level: an effective sample size $n$ and an effective event count $y$. There are two quite different ways to soften a disputed event.

SIX PRE-TREATED PATIENTS AT L1, ONE CONTESTED EVENT, HALF WEIGHT

| APPROACH | N | Y | WHAT IT ASSERTS |
|---|---|---|---|
| Block discount | 3.0 | 0.5 | All six observations are worth half as much |
| **Non-binary DLT** | 6.0 | 0.5 | All six were fully observed; half the event was real |

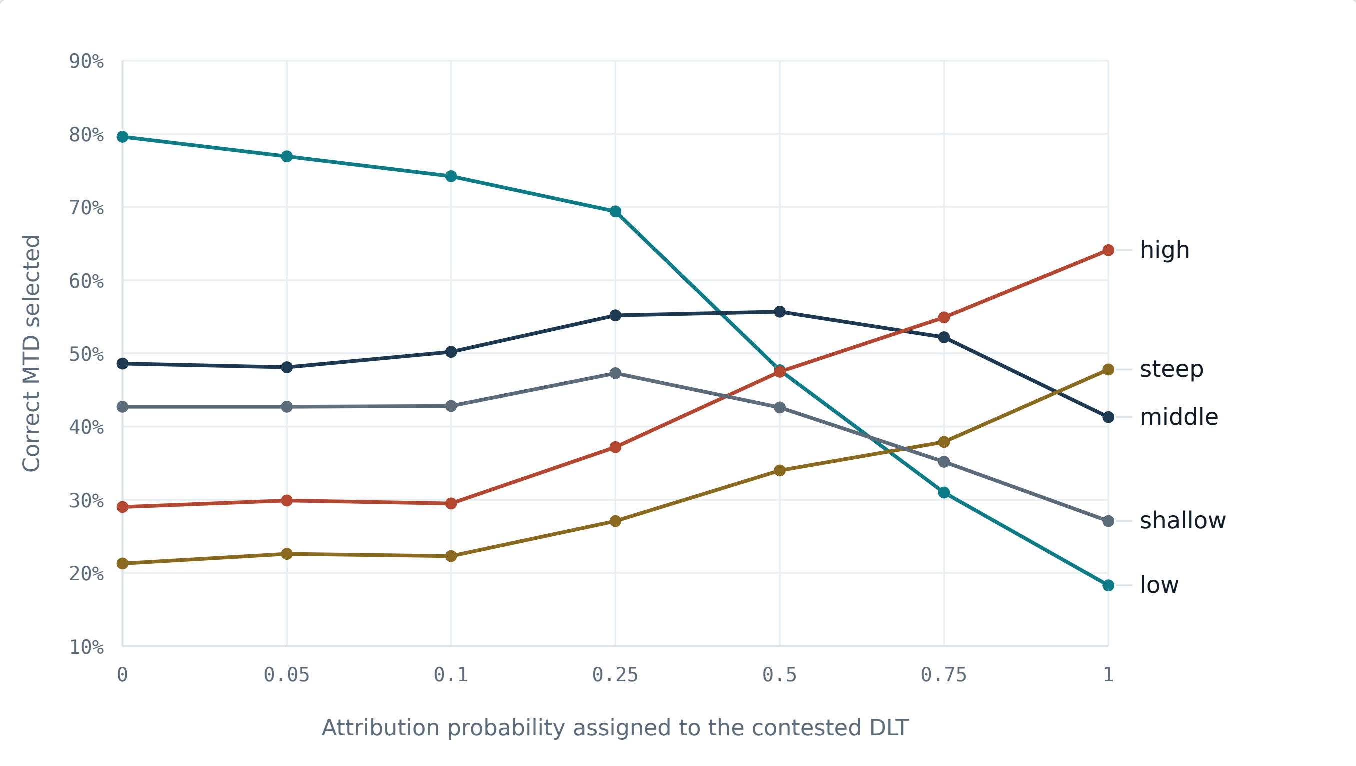
The block discount also halves the evidential weight of the **five patients who had no event at all** — patients about whom nothing is in doubt. Only the sixth is contested. Scaling $y$ alone isolates the uncertainty where it actually sits, and keeps the reassuring information the other five provide.

The two are not simply stronger and weaker versions of one another. Halving the block leaves the apparent toxicity rate at L1 unchanged at one in six, but makes the design less sure of it. Halving the event lowers the apparent rate itself, to half an event in six patients, while keeping the design just as sure. The same number therefore means something different in each scheme, and the two must be compared by what they buy, not by the label on the dial.

Compared that way — at equal risk of selecting an overly toxic dose — the non-binary DLT is worth 2.5 percentage points of accuracy on average (range 0.5 to 3.7). The advantage is real but moderate, and it comes from keeping the five event-free patients at full strength.

## 4 What the simulations show

We swept the attribution probability assigned to the contested event across 0, 0.05, 0.10, 0.25, 0.50 and 1.00, over all five acute toxicity scenarios, 1000 simulated trials per cell. At p = 1 the design is the conventional binary one; at p = 0 the event is never recorded.




**Correct MTD selection against attribution probability.** The scenarios move in opposite directions, which is the central finding. Where the true MTD is the top dose (low) a lower attribution helps sharply, from 18.3% at p = 1 to 79.6% at p = 0. Where the true MTD is low (high, steep) the same change hurts, from 64.1% down to 29.0%. There is no value of p that is best everywhere, because p encodes a belief about the world, not a tuning parameter.


**What each scheme buys, averaged over scenarios.** Both curves start from the same point at the lower left — the conventional binary design, which discounts nothing. As either dial is turned the design trades safety for accuracy, and the non-binary DLT sits above the block discount over the whole useful range: at equal risk it is worth 2.5 percentage points of accuracy on average. Note also that neither curve keeps climbing. Beyond roughly a third risk both turn over, so discounting the event away entirely is not the most accurate choice, merely the least safe.

CORRECT MTD SELECTION BY ATTRIBUTION PROBABILITY (%)

| P | LOW | MIDDLE | HIGH | STEEP | SHALLOW | AVERAGE | TOO HIGH (AVG) |
|---|---|---|---|---|---|---|---|
| 0 | 79.6 | 48.6 | 29.0 | 21.3 | 42.7 | **44.2** | 44.6 |
| 0.05 | 76.9 | 48.1 | 29.9 | 22.6 | 42.7 | **44.0** | 43.2 |
| 0.10 | 74.2 | 50.2 | 29.5 | 22.3 | 42.8 | **43.8** | 42.5 |
| 0.25 | 69.4 | 55.2 | 37.2 | 27.1 | 47.3 | **47.2** | 36.5 |
| **0.50** | 47.7 | 55.7 | 47.5 | 34.0 | 42.6 | **45.5** | 28.5 |
| 0.75 | 31.0 | 52.2 | 54.9 | 37.9 | 35.2 | **42.2** | 23.1 |
| 1.00 | 18.3 | 41.3 | 64.1 | 47.8 | 27.1 | **39.7** | 17.3 |

| NON-BINARY DLT AGAINST BLOCK DISCOUNT, AVERAGED OVER SCENARIOS | | | | | | |
|---|---|---|---|---|---|---|
| | NON-BINARY DLT | | BLOCK DISCOUNT | | NON-BINARY + OVERRIDE | |
| P | CORRECT | TOO HIGH | CORRECT | TOO HIGH | CORRECT | TOO HIGH |
| 0 | 44.2 | 44.6 | 43.6 | 38.5 | 43.8 | 45.5 |
| 0.05 | 44.0 | 43.2 | 44.2 | 37.6 | 45.0 | 43.1 |
| 0.10 | 43.8 | 42.5 | 43.3 | 35.4 | 44.8 | 42.6 |
| 0.25 | 47.2 | 36.5 | 42.7 | 32.0 | 46.5 | 37.7 |
| 0.50 | 45.5 | 28.5 | 42.1 | 26.1 | 44.2 | 30.1 |
| 0.75 | 42.2 | 23.1 | 40.6 | 20.5 | 43.3 | 24.2 |
| 1.00 | 39.7 | 17.3 | 39.9 | 17.1 | 41.9 | 18.3 |

The safety cost is monotone: averaged over scenarios, the risk of selecting an overly toxic MTD rises from 17.3% at $p = 1$ to 44.6% at $p = 0$. Nothing about making the endpoint continuous removes that trade-off. What it does is let the trade-off be set by a stated clinical belief rather than by a forced binary.

ACCURACY DOES NOT SIMPLY INCREASE AS THE EVENT IS DISCOUNTED

Averaged over the five scenarios, accuracy peaks at **p = 0.25** (47.2% correct at 36.5% risk) and falls away on both sides. Counting the event in full costs accuracy because it holds the design below a genuinely tolerable dose; discarding it entirely costs accuracy too, because the design then overshoots wherever the true MTD is low. A modest, stated discount does better than either extreme — which is a reasonable thing to find when the clinical assessment itself was neither 0 nor 1.

This should not be read as a recommendation to set $p = 0.25$. The average across five hypothetical scenarios is not a quantity anyone is trying to optimise, and p is meant to express a belief about this event, not to be tuned. It does show that the conventional binary choice is not the accuracy-maximising one under any reading.

## 5 Where radiotherapy may add something new

In the published attribution designs, p is a subjective judgement by the treating physician. That is the method's weak point, and the reason it has not been widely adopted: a number between 0 and 1 invites more variation between assessors than a binary does, and it is hard to audit.

Radiotherapy is unusual in offering an objective substrate for that judgement. For a given event one can ask where it occurred relative to the irradiated volume, what dose that volume received, and how that compares with the planning constraints. Our own serious adverse event documentation already reasons this way — recording that the stomach dose stayed within the predefined constraints, and undertaking to correlate the anatomical location of the finding with the treatment field.

That reasoning is currently written in prose and then collapsed into a binary. Turning it into a pre-specified, dosimetry-based mapping onto an attribution probability would be a genuine methodological contribution, and one that is specific to radiotherapy dose finding rather than borrowed from systemic oncology.

## 6 A practical elicitation scheme: discretize to the existing causality scale

### Asking a safety committee to name a continuous probability invites a question nobody can answer: why 0.35 and not 0.40? A coarser scheme avoids it, and one is already sitting on the SAE form.

Every serious adverse event is already classified on a five-tier causality scale — *unrelated, unlikely, possible, probable, definite* — the same scale used for MERGE-011. Rather than inventing a new categorical system, the committee could fix a representative attribution probability for each existing tier once, and apply that table case by case. That turns an unanswerable question about a decimal into a classification judgement clinicians already make routinely, and it reuses the discrete-score logic the quasi-CRM already applies to toxicity grade — the same mechanism, aimed at causality instead of severity.

SAE CAUSALITY TIER MAPPED TO A FIXED ATTRIBUTION PROBABILITY, AVERAGED OVER SCENARIOS

| CAUSALITY TIER | ATTRIBUTION P | CORRECT MTD | TOO HIGH |
|---|---|---|---|
| Unrelated | 0.00 | 44.2 | 44.6 |
| Unlikely | 0.05 | 44.0 | 43.2 |
| **Possible** | 0.25 | 47.2 | 36.5 |
| Probable | 0.75 | 42.2 | 23.1 |
| Definite | 1.00 | 39.7 | 17.3 |

Discretizing costs nothing in these simulations: the five tier values trace the same curve as the continuous sweep in section 4, because they are five points read off it. **MERGE-011 was assessed as “possible”** — which happens to be the tier with the best average accuracy, 47.2% correct at 36.5% risk, not because the analysis was built to reach that conclusion but because that is where the existing classification already places this event.

What this does not fix: it does not touch the quasi-likelihood issue in section 7, and it narrows rather than removes the incentive to shade an assessment, since a borderline case can still be pushed into a more favourable neighbouring tier. A visible jump between tiers is easier to question at review than an unexplained decimal, which is the actual gain — not the elimination of the underlying tension between speed and caution.

## 7 What would have to be settled first

1 · THE QUASI-LIKELIHOOD IS SHARPER THAN THE EVIDENCE

Substituting $y = p$ is not the same as marginalising over the unknown attribution. The correct mixture likelihood is $p \cdot P + (1-p)(1-P)$; the quasi-likelihood used here is $P^p(1-P)^{1-p}$. By Jensen's inequality the latter is the more confident of the two, so this implementation slightly overstates the information the disputed event carries. The published design handles this properly through data augmentation over the latent binary outcome, and a protocol version should do the same.

2 · WHO ASSIGNS P, AND WHEN

An assessor who knows the patient was treated at the top dose level may, without intending to, shade the number, and the incentive is sharper here than for an ordinary adverse-event grading: a lower attribution has an immediate, visible effect on the next dosing decision. Blinding assessment to dose level, the usual mitigation, is difficult to operationalise in a small radiotherapy trial where the treatment record makes the dose level obvious. Section 6 discusses a coarser elicitation scheme that narrows, without eliminating, this problem.

3 · SCOPE OF WHAT WAS SIMULATED

Only the one contested historical event was made fractional here. Every event arising during the simulated trial still counts as a full DLT. A design that applied attribution probabilities prospectively would also need a model of how assessors assign them, which is a substantially larger undertaking and would change these numbers.

4 · REGULATORY RECEPTION

A committee reading that a patient contributed 0.4 of a dose-limiting toxicity will reasonably ask what that means for the safety narrative. The argument is defensible — the graded assessment is already being made, and this stops discarding it — but it needs to be made explicitly rather than assumed.

## 8 Recommendation

FOR THE CURRENT AMENDMENT

**Do not adopt this now.** It is a larger methodological step than the amendment needs, it requires the statistician's input and a proper data-augmentation implementation, and it would complicate approval on a timeline aimed at April. Keep the amendment to the design transition and the weighting decision already set out.

Two things are nonetheless worth carrying forward immediately.

- **Correct the discounting we do adopt.** If the amendment down-weights the historical data at all, it should scale the event rather than the block. At equal risk this buys 2.5 percentage points of accuracy, and the argument for it is easier to make: we are uncertain about one event, not about five patients who had none. This change is small, self-contained, and does not require the full attribution machinery.
- **Pursue the dosimetric attribution mapping as a separate track.** The statistical machinery is published and citable; the radiotherapy-specific part is not, and that is where a real contribution would sit. This study is a well-documented motivating case for it.

### Suggested next step

Put the mixture-likelihood question to the trial statistician first. If a data-augmented implementation is straightforward in the existing R code, the remaining questions are clinical rather than statistical, and can be worked through by the team.

## 9 Scenarios and settings

True acute toxicity probabilities, with subacute probabilities fixed at 0.02, 0.05, 0.10, 0.15, 0.25. The true MTD is the highest level at or below the acute target of 0.20.

TRUE ACUTE TOXICITY PROBABILITIES BY SCENARIO

| SCENARIO | L0 5X4 GY | L1 5X5 GY | L2 5X6 GY | L3 5X7 GY | L4 5X8 GY | TRUE MTD |
|---|---|---|---|---|---|---|
| low | 0.01 | 0.02 | 0.06 | 0.10 | 0.15 | **L4** |
| middle | 0.01 | 0.04 | 0.12 | 0.17 | 0.27 | **L3** |
| high | 0.01 | 0.05 | 0.15 | 0.22 | 0.30 | **L2** |
| steep | 0.01 | 0.02 | 0.08 | 0.24 | 0.34 | **L2** |
| shallow | 0.10 | 0.13 | 0.15 | 0.18 | 0.24 | **L3** |

### References


- Yuan Z, Chappell R, Bailey H. The continual reassessment method for multiple toxicity grades: a Bayesian quasi-likelihood approach. *Biometrics* 2007.
- Phase I designs that allow for uncertainty in the attribution of adverse events. *Journal of the Royal Statistical Society Series C*.
- A Bayesian dose-finding design for outcomes evaluated with uncertainty. 2021.
- Quasi-partial order continual reassessment method: applying toxicity scores to cancer dose-finding drug combination trials. *Contemporary Clinical Trials* 2022.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Generated from `dlt_attribution_sweep.py` (1000 simulations per scenario per value, seed 20260813). TITE-CRM without EWOC and without burn-in, starting at L2, 30 patients including the six pre-treated, cohort size 3, one new patient every four weeks.